Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/07/2017
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/30/2013	**End Date:**	07/29/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/29/2016 per NSSC information (Ed., 7/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AK70G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Editor's note 11/11/2013. Note this is a continuation of "Space Radiation Effects on Genomic and Epigenetic Stability: Influence of Radiation Quality and Genetic Predisposition" (grant NNX11AC55G); see that project for previous reports.] Astronauts traveling in space are exposed to a variety of radiations. It is known that exposure to ionizing radiation can lead to an increased risk for developing cancer. Since there is currently no way to protect astronauts, it is important that the carcinogenic risk of exposures to these radiations is understood. The delayed, deleterious effects of radiation exposure are grouped under the terms radiation-induced genomic and epigenomic instability. These mechanisms are considered to be early steps in the carcinogenic process. Epigenetic mechanisms are thought to play a significant role in carcinogenesis by regulating gene expression and genome stability. These epigenetic mechanisms include DNA methylation, miRNA mediated gene silencing, and histone modification. There are also genetic factors in mouse and humans that determine the relative sensitivity of their cells to radiation-induced effects and ultimately their cancer risk. This project will develop assays that will provide functional biomarkers for exposure that are also indicators of carcinogenic risk. We will accomplish these objectives by evaluating mice with differing sensitivities to radiation for epigenomic instability as measured by changes in miRNA and DNA methylation profiles following exposure to iron, silicon, or oxygen ion radiations. The effect of age on the epigenetic radiation induced response will be evaluated using mice of relatively younger and older ages.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal of this grant was to evaluate the effect of space radiation exposure on epigenetic profiles with a focus on the role for genetic factors, age, and radiation linear energy transfer (LET). Some of the research and technology questions that this proposal addresses include the following: 1) the probabilities for increased carcinogenic risk following relatively low dose radiation exposure as a function of NASA's operational parameters (radiation quality, age at exposure, genetic factors, tissue specific response); 2) the development of quantitative, functional biomarkers for carcinogenic risk; 3) a better understanding of the molecular mechanisms involved in cancer initiation and progression. Understanding the cancer risk associated with exposure to either terrestrial or space radiation is important for human health. Epigenetic abnormalities are thought to play a key role in the earliest steps of carcinogenic initiation. Epigenetic changes are mitotically and meiotically heritable, stable alteration in gene expression that include DNA methylation, histone modifications, and RNA associated gene silencing. This project was designed to test the hypothesis that radiation exposure can affect epigenetic profiles and increase carcinogenic risk. We further hypothesize that genetic factors and age at the time of exposure impact the epigenetic radiation response. Specific Aim 1: evaluate radiation-induced alterations in miR profiles and gene expression. For each tissue, dose, and time point, genome-wide miR profiles will be analyzed. In parallel, global mRNA profiles will be analyzed to provide information regarding gene expression. Together these arrays will allow us to identify up-regulated miRNA that correlate with down-regulated target mRNA, or vis versa. Mining these data will allow us to identify affected pathways that are relevant to cancer and the radiation response. Specific aim 2: evaluate the effect of irradiation on the induction of aberrant DNA methylation. Alterations in DNA methylation have emerged as one of the most consistent molecular markers in multiple cancers and for carcinogenic initiation and progression. We will measure genome wide methylation by evaluating repeat element methylation. These repeat elements make up a significant portion of the mammalian genome and aberrant repeat element methylation has been linked to genomic instability, altered gene expression, and cancer. We will also evaluate promoter methylation for genes associated with increased carcinogenic risk and the radiation response. Using these three strategies to evaluate tissue specific DNA methylation in irradiated mice we will be able to survey genome wide changes that have been linked to carcinogenic processes. As a whole, this project proposed a systems biology approach to provide information regarding tissue-specific effects of radiation LET, genetic factors, age, and post-irradiation time on epigenetically-mediated gene regulation as it applies to the radiation response and radiation-induced carcinogenesis. This benefits not only our astronauts, but those of us on Earth, by increasing our understanding of the molecular mechanisms of cancer and the radiation response.
	During the past 3 years the project resulted in 6 publications. Three of those papers directly address epigenetic mechanisms involved in the radiation response. We showed a role for miR regulation of gene expression in radiation induced genomic instability (Baulch et al., 2014) and tissue specificity of DNA methylation profiles following high LET radiation exposure (Lima et al., 2013; Acharya et al., 2017 in press). Additional papers characterized 1) the types of DNA damage that persist in cells that exhibit radiation induced genomic instability (Keszenman et al., 2015), 2) the persistence of oxidative stress in human neural stem cells exposed to space radiations (Baulch et al., 2015), and 3) mechanisms by which space radiations induce cognitive dysfunction (Parihar et al., 2015, as well as Acharya et al., 2017 in press). Arguably the most important advances made during this project are those contained in our 2017 manuscript that is in press at Scientific Reports. The data presented in that paper demonstrate a clear correlation between epigenetic mechanisms and high LET radiation induced cognitive dysfunction (please find below the Abstract from that manuscript). Abstract (Acharya et al., 2017 manuscript that is in press at Scientific Reports): Among the dangers to astronauts engaging in deep space missions such as a Mars expedition is exposure to radiations that put them at risk for severe cognitive dysfunction. These radiation-induced cognitive impairments are accompanied by functional and structural changes including oxidative stress, neuroinflammation, and degradation of neuronal architecture. The molecular mechanisms that dictate CNS function are multifaceted and it is unclear how irradiation induces persistent alterations in the brain. Among those determinants of cognitive function are neuroepigenetic mechanisms that translate radiation responses into altered gene expression and cellular phenotype. In this study, we have demonstrated a correlation between epigenetic aberrations and adverse effects of space relevant irradiation on cognition. In cognitively impaired irradiated mice we observed increased 5-methylcytosine and 5-hydroxymethylcytosine levels in the hippocampus that coincided with increased levels of the DNA methylating enzymes DNMT3a, TET1, and TET3. By inhibiting DNA methylation using 5-iodotubercidine, we demonstrated amelioration of the epigenetic effects of irradiation. In addition to protecting against those molecular effects of irradiation, 5-iodotubercidine restored behavioral performance to that of unirradiated animals. The findings of this study establish the possibility that neuroepigenetic mechanisms significantly contribute to the functional and structural changes that affect the irradiated brain and cognition. Publications (see also Cumulative Bibliography): Baulch JE, Aypar U, Waters KM, Yang AJ, Morgan WF. "Genetic and epigenetic changes in chromosomally stable and unstable progeny of irradiated cells." PLoS One. 2014 Sep 24;9(9):e107722. eCollection 2014. <a target="_blank"

	href="http://dx.doi.org/10.1371/journal.pone.0107722">http://dx.doi.org/ ; PubMed PMID: 25251398; PubMed Central PMCID: PMC4175465
Task Progress:	Lima F, Ding D, Goetz W, Yang AJ, Baulch JE. "High LET 56Fe ion irradiation induces tissue-specific changes in DNA methylation in the mouse." Environmental and Molecular Mutagenesis. 2014 Apr;55(3):266-77. Epub 2013 Nov 30. http://dx.doi.org/ ; PubMed PMID: 24723241 Keszenman DJ, Kolodiuk L, Baulch JE. "DNA damage in cells exhibiting radiation-induced genomic instability." Mutagenesis. 2015 May;30(3):451-8. Epub 2015 Feb 22. http://dx.doi.org/ ; PubMed PMID: 25711497 Baulch JE, Craver BM, Tran KK, Yu L, Chmielewski N, Allen BD, Limoli CL. "Persistent oxidative stress in human neural stem cells exposed to low fluences of charged particles." Redox Biol. 2015 Aug;5:24-32. Epub 2015 Mar 11. http://dx.doi.org/ ; PubMed PMID: 25800120; PubMed Central PMCID: PMC4371546 Parihar VK, Allen B, Tran KK, Macaraeg TG, Chu EM, Kwok SF, Chmielewski NN, Craver BM, Baulch JE, Acharya MM, Cucinotta FA, Limoli CL. "What happens to your brain on the way to Mars." Sci Adv. 2015 May 1;1(4): e1400256. http://dx.doi.org/ Presentations from 2016-2017: January 2017 Human Research Program (HRP) meeting in Galveston TX. EPIGENETIC MECHANISMS OF THE CNS RESPONSE TO SPACE RADIATION EXPOSURE. Munjal M. Acharya, Al Anoud D. Baddour, Takumi Kawashita, Barrett D. Allen, Amber R. Syage, Thuan H. Nguyen, Nicole Yoon, Erich Giedzinski, Liping Yu, Vipan K. Parihar, and Janet E. Baulch. October 2016 Radiation Research Society meeting in Kona HI. Epigenetic determinants of space radiation induced cognitive dysfunction. Janet Baulch. February 2016 HRP meeting in Galveston TX. EPIGENETIC CHANGES IN THE BRAIN OF MICE EXPOSED TO 28SI PARTICLE RADIATION. M.M. Acharya, A.A.D. Baddour, T. Kowashita, N. Ru, J.E. Baulch.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Acharya MM, Baddour AAD, Kawashita T, Allen BD, Syage AR, Nguyen TH, Yoon N, Giedzinski E, Yu L, Parihar VK, Baulch JE. "Epigenetic determinants of space radiation-induced cognitive dysfunction." Scientific Reports. 2017 Feb 21;7:42885. http://dx.doi.org/10.1038/srep42885 ; PubMed PMID: 28220892; PubMed Central PMCID: PMC5318883 , Feb-2017
Articles in Peer-reviewed Journals	Parihar VK, Allen BD, Caressi C, Kwok S, Chu E, Tran KK, Chmielewski NN, Giedzinski E, Acharya MM, Britten RA, Baulch JE, Limoli CL. "Cosmic radiation exposure and persistent cognitive dysfunction." Sci Rep. 2016 Oct 10;6:34774. http://dx.doi.org/10.1038/srep34774 ; PubMed PMID: 27721383; PubMed Central PMCID: PMC5056393 , Oct-2016